

LLM-as-a-Judge Report: Repair-understanding GSM8K

What is measured?

Repair-understanding is the judge’s assessment of how much the model loses the intended meaning of the question because of the corrupted wording. A score of 0 means the meaning is fully understood; 5 means the model is fundamentally lost. This is not a label for whether the arithmetic answer is correct.

How to read the score

The judge uses a 0–5 scale. The reported value in the tables and graphs is the arithmetic mean over valid judged traces in the group. These are behavioral scores, not correctness scores: a model can be wrong without showing doubt, and it can be correct while expressing uncertainty.

Examples

The typo-percentage trend supports the requested direction: the pooled mean increases from 0.053 at 0% to 0.580 at 25%, 0.576 at 50%, and 0.764 at 75%. The real-word ratio comparison is less monotonic: it is 0.554 at 10%, 0.697 at 40%, and 0.669 at 70%. Thus more typo corruption is associated with lower understanding in this sample, while the real-word-ratio effect peaks at 40% rather than increasing through 70%.

Table 1: Examples from the per-configuration aggregate table. The relevant score is shown alongside both judge dimensions for context.

config	n	selected score	self-doubt	repair-understanding
clean	494	0.053	0.053	0.053
typo75_real10	482	0.660	0.660	0.660
typo75_real70	486	0.817	0.817	0.817

Aggregate result by typo percentage

Table 2: Mean score pooled by typo percentage. Clean is the 0% baseline; typo points pool all three real-word ratios.

typo %	n	mean score
0	494	0.053
25	1450	0.580
50	1454	0.576
75	1455	0.764

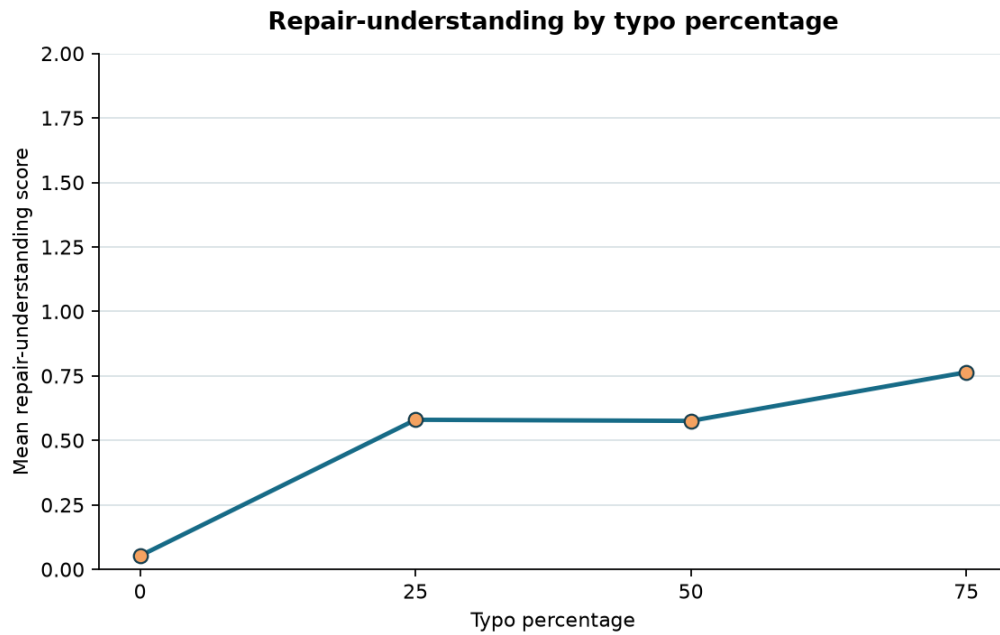


Figure 1: Repair-understanding pooled by typo percentage.

Aggregate result by real-word ratio

Table 3: Mean score pooled by real-word ratio. Each point pools the 25%, 50%, and 75% typo configurations.

real-word ratio %	n	mean score
10	1447	0.554
40	1457	0.697
70	1455	0.669

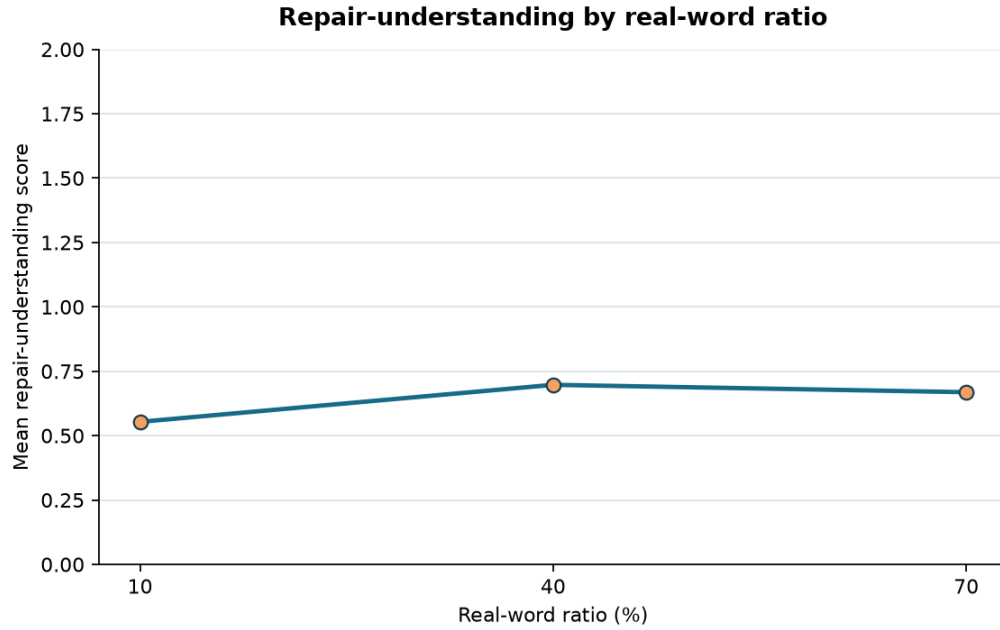


Figure 2: Repair-understanding pooled by real-word ratio.

Conclusion

The typo-percentage trend supports the requested direction: the pooled mean increases from 0.053 at 0% to 0.580 at 25%, 0.576 at 50%, and 0.764 at 75%. The real-word ratio comparison is less monotonic: it is 0.554 at 10%, 0.697 at 40%, and 0.669 at 70%. Thus more typo corruption is associated with lower understanding in this sample, while the real-word-ratio effect peaks at 40% rather than increasing through 70%.